

Database Design Project

Oracle Baseball League Store Database

Project Scenario:

You are a small consulting company specializing in database development. You have just been awarded the contract to develop a data model for a database application system for a small retail store called Oracle Baseball League (OBL).

The Oracle Baseball League store serves the entire surrounding community selling baseball kit. The OBL has two types of customer, there are individuals who purchase items like balls, cleats, gloves, shirts, screen printed t-shirts, and shorts. Additionally customers can represent a team when they purchase uniforms and equipment on behalf of the team.

Teams and individual customers are free to purchase any item from the inventory list, but teams get a discount on the list price depending on the number of players. When a customer places an order we record the order items for that order in our database.

OBL has a team of three sales representatives that officially only call on teams but have been known to handle individual customer complaints.

Section 3 Lesson 4 Exercise : Data Modeling Terminology and Mapping

Apply the Rules of Relationship Mapping to Transform Relationship (S3L4 Objective 3)

Part 1 : Create Table Mapping Document

Data modeling allows you to understand and apply the Oracle naming conventions of a Conceptual Data Model and apply them in the creation of a Physical Data Model. The Physical model will represent the information stored in the ERD in a way that a relational database can be easily produced.

Using the ERD that you have produced throughout this project (or the suggested start point in this document) create a table mapping document that will represent the transformation of the Conceptual Data Model terminology to that of the Physical Data Model.

Table Mapping instructions

1. The first row of the table diagram contains the table name and the short name.
 - a. The table name is the plural of the entity name.
 - b. The table short name is what will be used to name the foreign key relationships and should be created following the appropriate strategy for the name of the entity.
2. The Key Type column should contain values of "pk" for the primary key, "uk" for the unique key, or "fk" for the foreign key column. The cell is blank if the column is not a part of a key.

When mapping foreign key attributes you should remember the following:

- a. You create the foreign key on the side of the relationship that has a crow's foot notation.
 - b. You take the primary key from the related table and add it using the table short name_attribute name as the identifier.
 - c. There is no specific order for the foreign keys to be identified by, but we number their type to show that there is more than one foreign key.
 - d. On a 1:1 relationship the foreign key can be added to the most appropriate table.
3. The Optionality column must contain an asterisk (*) if the column is mandatory and a lowercase "o" if it is optional.
4. The third column is for the column name this should match the singular attribute name but with spaces replaced with underscores. The name of the column should not include the entity/table name because columns are qualified with the table name
5. Remember that you have to follow the naming conventions at all stages and you have to be consistent in your naming approach.
6. Be careful not to try to assign keywords as table/column names as this will cause you problems when you try to create the Physical model.

The first table (customers) has been completed for you as an example.

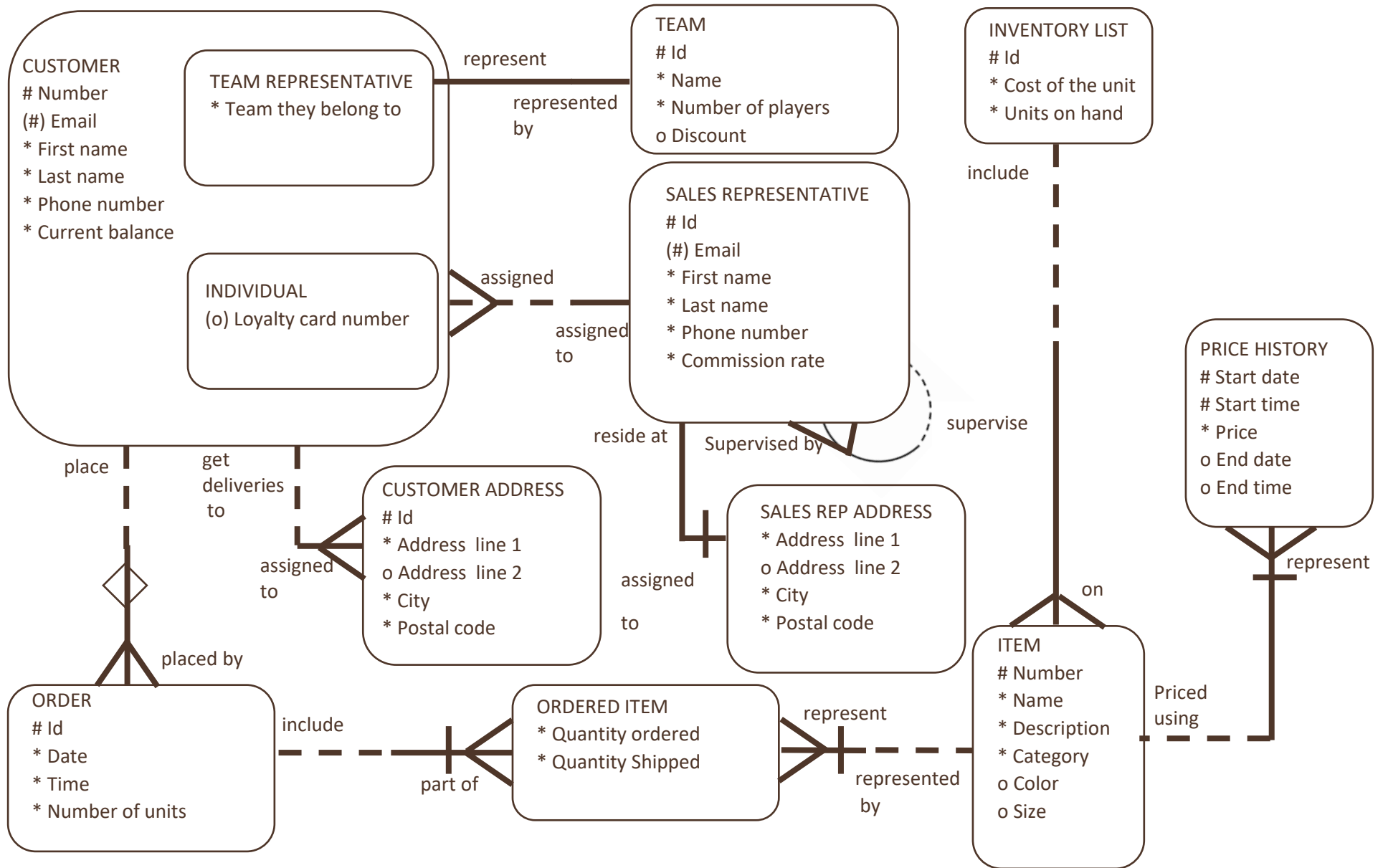
A single table implementation has been used to represent the super/subtypes shown in the CUSTOMER entity.

Table Name	Table Short Name	
customers	ctr	
Key Type	Optionality	Column Name
pk	*	ctr_number
uk	*	email
	*	first_name
	*	last_name
	*	phone_number
	*	current_balance
uk	o	loyalty_card_number
fk1	o	tem_id
fk2	o	sre_id

Using the given example and the following template to complete a separate table for every entity in your Conceptual Data Model.

Table Name	Table Short Name	
Key Type	Optionality	Column Name

Suggested starting point



Part 2 : Modifying Table Mapping Document

Data modeling allows you to understand and apply the Oracle naming conventions of a Conceptual Data Model and apply them in the creation of a Physical Data Model. The Physical model will represent the information stored in the ERD in a way that a relational database can be easily produced.

Using the table mapping document that you created in part 1 of this exercise complete the tables to also show appropriate data types and sizes.

Table Mapping instructions

1. The provided suggested starting point in this document shows you the relationships between the tables with sample data for each field.
2. The colour coding shows you where the primary key/foreign key relationships occur.
3. The columns that hold the foreign key values need to be of the same data type (and preferably size) in order to hold that data that will forge the relationship.
4. The data provided is only a sample and the content shouldn't be taken as a maximum size.
5. For each field calculate the maximum size that you think is the most appropriate for the type of information that will be stored there.

The first table (customers) has been completed for you as an example.

A single table implementation has been used to represent the super/subtypes shown in the CUSTOMER entity.

Table Name	Table Short Name			
customers	ctr			
Key Type	Optionality	Column Name	Data type	size
pk	*	ctr_number	VARCHAR2	6
uk	*	email	VARCHAR2	50
	*	first_name	VARCHAR2	20
	*	last_name	VARCHAR2	30
	*	phone_number	VARCHAR2	11
	*	current_balance	NUMBER	6,2
uk	o	loyalty_card_number	VARCHAR2	6
fk1	o	tem_id	VARCHAR2	4
fk2	o	sre_id	VARCHAR2	4

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Using the given example and the following template complete a separate table for every entity in your Conceptual Data Model that will show the data type and size (if appropriate) for each attribute..

Table Name	Table Short Name			
Key Type	Optionality	Column Name	Data type	size

Suggested starting point:

Sample Table Data - The following tables represent the first row of data in each table you should place the data types and sizes in the appropriate table that you created in Task 1.

customers		ctr						
ctr_number	email	first_name	last_name	Phone_number	current_balance	loyalty_card_number	tem_id	sre_id
c00001	bob.thornberry@heatmail.com	Robert	Thornberry	01234567898	150.00		t001	sr001

customers_addresses					
id	address_line_1	address_line_2	city	postal_code	ctr_number
ca0101	83 Barrhill Drive		Liverpool	Lp79HJK	c00001

teams

id	name	number_of_players	discount
t001	Rockets	25	10

sales_representatives

id	email	first_name	last_name	phone_number	commission_rate	supervisor_id
sr01	chray@obl.com	Charles	Raymond	134598761	5	sr01

sales_representatives_address

id	address_line_1	address_line_2	city	postal_code
sr01	12 Cherry Lane	Denton	Detroit	DT48211

orders

odr

id	date	time	number_of_units	ctr_number
or0101250	17-Jun-2019	08:32:30	10	c00001

ordered_items

oim

quantity_ordered	quantity_shipped	odr_id	itm_number
5	5	or0101250	lm01101025

Items

itm

itm_number	name	description	category	color	size	ilt_id
lm01101025	gloves	catcher mitt	clothing	brown	m	il010230124

Inventory_list

ilt

id	unit_cost	units_on_hand
il010230124	2.50	100

price_history

start_date	start_time	price	end_date	end_time	itm_number
17-Jun-2019	09:00:00	4.99			lm01101025